

Universidad de Costa Rica

Facultad de Ingeniería

Escuela de Ciencias de la Computación e Informática

CI-0163 - Análisis de Grandes Volúmenes de Datos

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Grupo 01

Laboratorio #3: OpenRefine

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Utilizamos el dataset Smartwatch Health Data, ya que este tiene datos interesantes y cumple con los requisitos de necesitar ser limpiado

Antes

★	1.	4174	58.93977586209927	98.80965025422655	5450.390577516058	7.167235622316564	Highly Active	1
★				98.53219515895226	727.6016100461258	6.538239375570314	Highly_Active	5
★	2.	1860	247.80305229234318	97.05295352461383	2826.5219939131794	ERROR	Highly Active	5
★	3.	2294	40.0	96.89421334689396	13797.338044243397	7.367789630207228	Active	3
★	4.	2130	61.950165287154064	98.58379675916517	15679.067648456179		Highly_Active	6
★	5.	2095	96.28593768588723	94.2029097779196	10205.992256276271	8.378342673824589	Highly_Active	10
★	6.	4772	47.27225670219411	95.38976001421	3208.781176562847	7.871146008904113	Sedentary	2
★	7.	4092	81.73349679221296	95.98134260868015	6051.249856733786	5.224139066195455	Sedentary	1
★	8.	3979	93.34033768144009	97.21894484728315	1775.0295780773379	5.468413771417153	Sedentary	1
★	9.	3169	51.40996727987549	97.27278686705215	7940.021451995014	7.9544989905339145	Sedentary	8
★	10.	1466	101.91493806087537		2987.568640188662	7.541956463173237	Sedentary	9
★	11.	2238	73.85294067228806	95.39944002358109	1748.1970688253557	4.830161671840198	Active	1
★	12.	1330	95.37229036722503	99.15265159443172	6155.837341884333	5.814348465914493	Highly_Active	10
★	13.	2482	79.06263167243848	99.01491345546357	829.9256232943183	6.269060283034982	Sedentary	
★	14.	3135	68.09020292958444	99.72973954402643	8124.135791328726	6.444869604525839	Highly_Active	5
★	15.	4444	40.0	99.04772547056064	1449.2158952991972	6.392173806051382	Active	2
★	16.	4171	58.2779626987928	97.78972645147674	5643.511357828403	7.848859015777246	Sedentary	10
★	17.	3919		96.05285348429064	3488.9195075738367	6.718661172525927	Sedentary	4
★	18.	4735	98.05924946042926	96.02709912891096	5846.923275235294	6.621126995655419	Sedentary	9
★			53.93020624079926	96.30256923405778	1976.1863916271682	5.3648491527683895	Highly_Active	5
★	19.	2685	74.10500253274513	99.64606450575437	2601.3964953506083	6.751333320225251	Highly Active	9
★	20.	4380	68.31010413084853		1058.3742744802673	6.456918401449286	Active	3
★	21.	1769	81.26723472758928	96.06816145748866	10503.006242728434	5.780212383241095	Active	
★	22.	3391	76.32552029631407	98.75711035080639	508.8380975267676	4.773999479227033	Sedentary	Very High
★	23.	2515	69.42479491992782	99.25495076372934	3306.1104116230244	8.41542161341735	Highly Active	1
★	24.	4485	84.58662488247226	99.02468690832053	4173.373029062226	9.169583797303172	Sedentary	9
★	25.	3853	64.62704386367523	99.78959923982669	1520.184386862472	6.866753232803355	Sedentary	2
★	26.	3433	75.16860350791544	98.10010128415522	3400.900980332703	8.651900418719752	Sedentary	10
★	27.	2215	86.5515986753003	97.47263459341158	3790.054977605134	6.844302018581762	Sedentary	1

Se utilizó el comando GREL

```
if(value == "Very High", 10, value)
```

para que cambie los valores donde dice Very High a 10

También se utilizó

```
if(value == "ERROR", "", value)
```

para cambiar los valores que dicen ERROR a un vacío

Para el valor del nivel de estrés y actividad, se utilizaron facets para conseguir la moda y posteriormente guardar dichos datos en los valores vacíos.

De igual forma, para rellenar los valores con valores vacíos en ritmo cardiaco, nivel de oxígeno en sangre, contador de pasos y horas de sueño, se descargó el documento formato excel, se sacó la mediana y se reemplazaron los valores vacíos por dicha mediana.

Redondeo de valores numéricos a 2 decimales con GREL:

```
value.toNumber().toString().slice(0, value.toString().indexOf(".") + 3).toNumber()
```

Resultado

☆	1.	1001	63.81	97.11	4552.29	4.47	Highly Active	4
☆	2.	1002	55.98	99.11	5334.01	9.04	Sedentary	6
☆	3.	1003	66.96	100	13053.92	6.48	Active	7
☆	4.	1004	79.4	98.02	3842.66	8.06	Active	8
☆	5.	1005	70.6	98.11	12133.01	5.54	Highly Active	6
☆	6.	1006	86.18	95.33	7085.29	9.31	Sedentary	7
☆	7.	1007	52.97	97.89	4170.4	5.72	Active	10
☆	8.	1009	84.35	99.52	5310.26	5.95	Sedentary	2
☆	9.	1010	99.83	97.06	5237.75	6.17	Highly Active	2
☆	10.	1011	50.6	97.44	12131.98	6.68	Highly Active	1
☆	11.	1012	48.91	97.91	9246.52	8.53	Active	9
☆	12.	1013	74.6	99.06	3019.24	3.97	Active	5
☆	13.	1015	78.61	97.44	6014.62	6.87	Highly Active	10
☆	14.	1016	80.62	100	13900.01	5.24	Sedentary	1
☆	15.	1018	67.46	97.24	1943.92	7.06	Highly Active	5
☆	16.	1019	57.23	96.43	2701.28	9.06	Highly Active	1
☆	17.	1020	54.16	97.34	5657.93	5.22	Sedentary	2
☆	18.	1021	67.22	100	6913.14	5.73	Active	10
☆	19.	1022	74.86	95.75	8776.47	6.5	Sedentary	7
☆	20.	1024	62.37	98.29	13302.57	6.15	Highly Active	5
☆	21.	1025	58.68	100	18958.71	7.71	Active	9
☆	22.	1026	75.06	99.57	9502.86	10.1	Highly Active	9
☆	23.	1027	75.06	95.99	7015.42	6.9	Highly Active	7
☆	24.	1028	79.75	98.46	19047.9	6.03	Highly Active	1
☆	25.	1029	83	100	5973.91	6.85	Active	2
☆	26.	1030	62.23	96.83	979.4	3.94	Highly Active	5
☆	27.	1032	52.72	99.85	17046.7	6.75	Active	1
☆	28.	1034	87.2	97.6	12948.3	6.14	Sedentary	5
☆	29.	1035	64.89	100	3162	6.97	Active	5